

Github: <https://github.com/movchun567/apz-kravchuk.git> , branch micro_hazelcast

Запускаємо docker compose і перевіряємо чи всі компоненти піднялись:

```
akravchuk@macA apz-kravchuk % docker compose ps
NAME                                IMAGE                                COMMAND                                SERVICE    CREATED   STATUS    PORTS
apz-kravchuk-counter-service-1      apz-kravchuk-counter-service      "gunicorn -w 1 -b 0..."            counter-service  4 minutes ago  Up 4 minutes  0.0.0.0:8002->8002/tcp
apz-kravchuk-facade-service-1       apz-kravchuk-facade-service       "gunicorn -w 1 -b 0..."            facade-service   4 minutes ago  Up 4 minutes  0.0.0.0:8000->8000/tcp
apz-kravchuk-hazelcast1-1           hazelcast/hazelcast:5.3           "hz start"                           hazelcast1       4 minutes ago  Up 4 minutes  5701/tcp
apz-kravchuk-hazelcast2-1           hazelcast/hazelcast:5.3           "hz start"                           hazelcast2       4 minutes ago  Up 4 minutes  5701/tcp
apz-kravchuk-hazelcast3-1           hazelcast/hazelcast:5.3           "hz start"                           hazelcast3       4 minutes ago  Up 4 minutes  5701/tcp
apz-kravchuk-logging-service-1-1    apz-kravchuk-logging-service-1    "gunicorn -w 1 -b 0..."            logging-service-1 4 minutes ago  Up 4 minutes  0.0.0.0:8001->8001/tcp
apz-kravchuk-logging-service-2-1    apz-kravchuk-logging-service-2    "gunicorn -w 1 -b 0..."            logging-service-2 4 minutes ago  Up 4 minutes  0.0.0.0:8003->8001/tcp
apz-kravchuk-logging-service-3-1    apz-kravchuk-logging-service-3    "gunicorn -w 1 -b 0..."            logging-service-3 4 minutes ago  Up 4 minutes  0.0.0.0:8004->8001/tcp
apz-kravchuk-postgres-1             postgres:16-alpine                "docker-entrypoint.s..."           postgres        4 minutes ago  Up 4 minutes  5432/tcp
```

Як бачимо, всі компоненти піднялись, тому тепер виконаємо тести.

10 транзакцій та логи сервісів:

The screenshot shows a REST client interface with the following details:

- URL:** `http://localhost:8000/transaction`
- Method:** `POST`
- Body:**

```
{
  "user_id": "user1",
  "amount": 1,
  "transaction_id": "msg10",
  "message": "msg10"
}
```
- Response:**

```
{
  "balance": 10,
  "transaction_id": "msg10"
}
```



http://localhost:8000/user/user1

GET



http://localhost:8000/user/user1

Docs

Params

Authorization

Headers (9)

Body ●

Scripts

Settings

Body

Cookies (1)

Headers (5)

Test Results



{ } JSON

Preview

Visualize



```
1 {
2   "balance": 10,
3   "transactions": [
4     {
5       "amount": 1,
6       "message": "msg1",
7       "timestamp": 1774544741.9305894,
8       "transaction_id": "msg1",
9       "user_id": "user1"
10    },
11    {
12      "amount": 1,
13      "message": "msg2",
14      "timestamp": 1774544754.6602273,
15      "transaction_id": "msg2",
16      "user_id": "user1"
17    },
18    {
19      "amount": 1,
20      "message": "msg3",
21      "timestamp": 1774544785.0147805,
22      "transaction_id": "msg3",
23      "user_id": "user1"
24    },
25    {
26      "amount": 1,
27      "message": "msg4",
28      "timestamp": 1774544792.1853397,
29      "transaction_id": "msg4",
30      "user_id": "user1"
31    },
32    {
33      "amount": 1,
34      "message": "msg5",
35      "timestamp": 1774544799.3685262,
36      "transaction_id": "msg5",
37      "user_id": "user1"
38    }
39  ]
40 }
```

Бачимо що транзакції виконано, тепер перевіримо логи кожного logging service:

```
✓ container apz-kravchuk-facade-service-1 Started
akravchuk@macA apz-kravchuk % docker logs apz-kravchuk-logging-service-1-1

[2026-03-26 17:05:19 +0000] [1] [INFO] Starting gunicorn 22.0.0
[2026-03-26 17:05:19 +0000] [1] [INFO] Listening at: http://0.0.0.0:8001 (1)
[2026-03-26 17:05:19 +0000] [1] [INFO] Using worker: sync
[2026-03-26 17:05:19 +0000] [7] [INFO] Booting worker with pid: 7
[logging-service:1] stored tx=msg2 user=user1 amount=1 msg=msg2
172.18.0.10 - - [26/Mar/2026:17:05:54 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[logging-service:1] stored tx=msg5 user=user1 amount=1 msg=msg5
172.18.0.10 - - [26/Mar/2026:17:06:39 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[logging-service:1] stored tx=msg9 user=user1 amount=1 msg=msg9
172.18.0.10 - - [26/Mar/2026:17:07:11 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[logging-service:1] stored tx=msg10 user=user1 amount=1 msg=msg10
172.18.0.10 - - [26/Mar/2026:17:07:18 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
akravchuk@macA apz-kravchuk % docker logs apz-kravchuk-logging-service-2-1

[2026-03-26 17:05:19 +0000] [1] [INFO] Starting gunicorn 22.0.0
[2026-03-26 17:05:19 +0000] [1] [INFO] Listening at: http://0.0.0.0:8001 (1)
[2026-03-26 17:05:19 +0000] [1] [INFO] Using worker: sync
[2026-03-26 17:05:19 +0000] [7] [INFO] Booting worker with pid: 7
[logging-service:2] stored tx=msg1 user=user1 amount=1 msg=msg1
172.18.0.10 - - [26/Mar/2026:17:05:42 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[logging-service:2] stored tx=msg3 user=user1 amount=1 msg=msg3
172.18.0.10 - - [26/Mar/2026:17:06:25 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[logging-service:2] stored tx=msg4 user=user1 amount=1 msg=msg4
172.18.0.10 - - [26/Mar/2026:17:06:32 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[logging-service:2] stored tx=msg6 user=user1 amount=1 msg=msg6
172.18.0.10 - - [26/Mar/2026:17:06:50 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[logging-service:2] stored tx=msg7 user=user1 amount=1 msg=msg7
172.18.0.10 - - [26/Mar/2026:17:06:57 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[logging-service:2] stored tx=msg8 user=user1 amount=1 msg=msg8
172.18.0.10 - - [26/Mar/2026:17:07:04 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
akravchuk@macA apz-kravchuk % docker logs apz-kravchuk-logging-service-3-1

[2026-03-26 17:05:19 +0000] [1] [INFO] Starting gunicorn 22.0.0
[2026-03-26 17:05:19 +0000] [1] [INFO] Listening at: http://0.0.0.0:8001 (1)
[2026-03-26 17:05:19 +0000] [1] [INFO] Using worker: sync
[2026-03-26 17:05:19 +0000] [7] [INFO] Booting worker with pid: 7
akravchuk@macA apz-kravchuk %
```

Бачимо, що сервіс 1 опрацював транзакції 2, 5, 9 та 10, а сервіс 2 всі інші. Сервіс 3 нічого не опрацював.

Далі пробуємо виключити один з logging service(а саме сервіс 2) і перевіряємо чи він вимкнувся:

```
akravchuk@macA apz-kravchuk % docker compose ps
NAME                                IMAGE                                COMMAND                                SERVICE    CREATED    STATUS    PORTS
apz-kravchuk-counter-service-1      apz-kravchuk-counter-service       "gunicorn -w 1 -b 0..."            counter-service  21 minutes ago    Up 21 minutes    0.0.0.0:8002->8002/tcp
apz-kravchuk-facade-service-1       apz-kravchuk-facade-service        "gunicorn -w 1 -b 0..."            facade-service  21 minutes ago    Up 21 minutes    0.0.0.0:8000->8000/tcp
apz-kravchuk-hazelcast1-1           hazelcast/hazelcast:5.3            "hz start"                            hazelcast1      21 minutes ago    Up 21 minutes    5701/tcp
apz-kravchuk-hazelcast2-1           hazelcast/hazelcast:5.3            "hz start"                            hazelcast2      21 minutes ago    Up 21 minutes    5701/tcp
apz-kravchuk-hazelcast3-1           hazelcast/hazelcast:5.3            "hz start"                            hazelcast3      21 minutes ago    Up 21 minutes    5701/tcp
apz-kravchuk-logging-service-1-1    apz-kravchuk-logging-service-1     "gunicorn -w 1 -b 0..."            logging-service-1  21 minutes ago    Up 21 minutes    0.0.0.0:8001->8001/tcp
apz-kravchuk-logging-service-3-1    apz-kravchuk-logging-service-3     "gunicorn -w 1 -b 0..."            logging-service-3  21 minutes ago    Up 21 minutes    0.0.0.0:8004->8001/tcp
apz-kravchuk-postgres-1             postgres:16-alpine                  "docker-entrypoint.s..."            postgres        21 minutes ago    Up 21 minutes    5432/tcp
```

Тепер можемо виконати ще 6 транзакцій і перевірити логи:

HTTP http://localhost:8000/user/user1

GET http://localhost:8000/user/user1

Docs Params Authorization Headers (9) Body ● Scripts Settings

Body Cookies (1) Headers (5) Test Results

{ } JSON Preview Visualize

```
77         "timestamp": 1774546113.2234118,\n78         "transaction_id": "msg11",\n79         "user_id": "user1"\n80     },\n81     {\n82         "amount": 1,\n83         "message": "msg12",\n84         "timestamp": 1774546113.2234118,\n85         "transaction_id": "msg12",\n86         "user_id": "user1"\n87     },\n88     {\n89         "amount": 1,\n90         "message": "msg13",\n91         "timestamp": 1774546117.588149,\n92         "transaction_id": "msg13",\n93         "user_id": "user1"\n94     },\n95     {\n96         "amount": 1,\n97         "message": "msg14",\n98         "timestamp": 1774546123.396949,\n99         "transaction_id": "msg14",\n100        "user_id": "user1"\n101    },\n102    {\n103        "amount": 1,\n104        "message": "msg15",\n105        "timestamp": 1774546128.7262225,\n106        "transaction_id": "msg15",\n107        "user_id": "user1"\n108    },\n109    {\n110        "amount": 1,\n111        "message": "msg16",\n112        "timestamp": 1774546136.074012,\n113        "transaction_id": "msg16",\n114        "user_id": "user1"
```

Тепер перевіряємо логи:

```

akravchuk@macA apz-kravchuk % docker logs apz-kravchuk-logging-service-1-1

[2026-03-26 17:05:19 +0000] [1] [INFO] Starting gunicorn 22.0.0
[2026-03-26 17:05:19 +0000] [1] [INFO] Listening at: http://0.0.0.0:8001 (1)
[2026-03-26 17:05:19 +0000] [1] [INFO] Using worker: sync
[2026-03-26 17:05:19 +0000] [7] [INFO] Booting worker with pid: 7
[logging-service:1] stored tx=msg2 user=user1 amount=1 msg=msg2
172.18.0.10 - - [26/Mar/2026:17:05:54 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[logging-service:1] stored tx=msg5 user=user1 amount=1 msg=msg5
172.18.0.10 - - [26/Mar/2026:17:06:39 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[logging-service:1] stored tx=msg9 user=user1 amount=1 msg=msg9
172.18.0.10 - - [26/Mar/2026:17:07:11 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[logging-service:1] stored tx=msg10 user=user1 amount=1 msg=msg10
172.18.0.10 - - [26/Mar/2026:17:07:18 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
172.18.0.10 - - [26/Mar/2026:17:09:04 +0000] "GET /transactions/user/user1 HTTP/1.1" 200 1049 "-" "python-requests/2.32.3"
[logging-service:1] stored tx=msg12 user=user1 amount=1 msg=msg12
172.18.0.10 - - [26/Mar/2026:17:28:33 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
akravchuk@macA apz-kravchuk % docker logs apz-kravchuk-logging-service-2-1

[2026-03-26 17:05:19 +0000] [1] [INFO] Starting gunicorn 22.0.0
[2026-03-26 17:05:19 +0000] [1] [INFO] Listening at: http://0.0.0.0:8001 (1)
[2026-03-26 17:05:19 +0000] [1] [INFO] Using worker: sync
[2026-03-26 17:05:19 +0000] [7] [INFO] Booting worker with pid: 7
[logging-service:2] stored tx=msg1 user=user1 amount=1 msg=msg1
172.18.0.10 - - [26/Mar/2026:17:05:42 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[logging-service:2] stored tx=msg3 user=user1 amount=1 msg=msg3
172.18.0.10 - - [26/Mar/2026:17:06:25 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[logging-service:2] stored tx=msg4 user=user1 amount=1 msg=msg4
172.18.0.10 - - [26/Mar/2026:17:06:32 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[logging-service:2] stored tx=msg6 user=user1 amount=1 msg=msg6
172.18.0.10 - - [26/Mar/2026:17:06:50 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[logging-service:2] stored tx=msg7 user=user1 amount=1 msg=msg7
172.18.0.10 - - [26/Mar/2026:17:06:57 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[logging-service:2] stored tx=msg8 user=user1 amount=1 msg=msg8
172.18.0.10 - - [26/Mar/2026:17:07:04 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[2026-03-26 17:26:55 +0000] [1] [INFO] Handling signal: term
[2026-03-26 17:26:55 +0000] [7] [INFO] Worker exiting (pid: 7)
[2026-03-26 17:26:55 +0000] [1] [INFO] Shutting down: Master
akravchuk@macA apz-kravchuk % docker logs apz-kravchuk-logging-service-3-1

[2026-03-26 17:05:19 +0000] [1] [INFO] Starting gunicorn 22.0.0
[2026-03-26 17:05:19 +0000] [1] [INFO] Listening at: http://0.0.0.0:8001 (1)
[2026-03-26 17:05:19 +0000] [1] [INFO] Using worker: sync
[2026-03-26 17:05:19 +0000] [7] [INFO] Booting worker with pid: 7
[logging-service:3] stored tx=msg11 user=user1 amount=1 msg=msg11
172.18.0.10 - - [26/Mar/2026:17:28:27 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[logging-service:3] stored tx=msg13 user=user1 amount=1 msg=msg13
172.18.0.10 - - [26/Mar/2026:17:28:37 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[logging-service:3] stored tx=msg14 user=user1 amount=1 msg=msg14
172.18.0.10 - - [26/Mar/2026:17:28:43 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[logging-service:3] stored tx=msg15 user=user1 amount=1 msg=msg15
172.18.0.10 - - [26/Mar/2026:17:28:48 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[logging-service:3] stored tx=msg16 user=user1 amount=1 msg=msg16
172.18.0.10 - - [26/Mar/2026:17:28:56 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
akravchuk@macA apz-kravchuk %

```

Бачимо що сервіс 2 вимкнувся, сервіс 1 отримав транзакцію 11, а сервіс 3 інші п'ять транзакцій. Вмикаємо сервіс 2 назад.

Тепер тестуємо що буде якщо вимкнути одну hazelcast ноду. Вимикаємо ноду 2 і перевіряємо:

```

akravchuk@macA apz-kravchuk % docker compose ps

```

NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS	PORTS
apz-kravchuk-counter-service-1	apz-kravchuk-counter-service	"gunicorn -w 1 -b 0..."	counter-service	35 minutes ago	Up 35 minutes	0.0.0.0:8002->8002/tcp
apz-kravchuk-facade-service-1	apz-kravchuk-facade-service	"gunicorn -w 1 -b 0..."	facade-service	35 minutes ago	Up 35 minutes	0.0.0.0:8000->8000/tcp
apz-kravchuk-hazelcast1-1	hazelcast/hazelcast:5.3	"hz start"	hazelcast1	35 minutes ago	Up 35 minutes	5701/tcp
apz-kravchuk-hazelcast3-1	hazelcast/hazelcast:5.3	"hz start"	hazelcast3	35 minutes ago	Up 35 minutes	5701/tcp
apz-kravchuk-logging-service-1-1	apz-kravchuk-logging-service-1	"gunicorn -w 1 -b 0..."	logging-service-1	35 minutes ago	Up 35 minutes	0.0.0.0:8001->8001/tcp
apz-kravchuk-logging-service-2-1	apz-kravchuk-logging-service-2	"gunicorn -w 1 -b 0..."	logging-service-2	35 minutes ago	Up 4 minutes	0.0.0.0:8003->8001/tcp
apz-kravchuk-logging-service-3-1	apz-kravchuk-logging-service-3	"gunicorn -w 1 -b 0..."	logging-service-3	35 minutes ago	Up 35 minutes	0.0.0.0:8004->8001/tcp
apz-kravchuk-postgres-1	postgres:16-alpine	"docker-entrypoint.s..."	postgres	35 minutes ago	Up 35 minutes	5432/tcp

Далі робимо ще 6 транзакцій:

HTTP <http://localhost:8000/user/user1>

GET <http://localhost:8000/user/user1>

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Query Params

Key
Key

Body Cookies (1) Headers (5) Test Results

{ } JSON Preview Visualize

```
137 {
138   "amount": 1,
139   "message": "msg20",
140   "timestamp": 1774546783.3151305,
141   "transaction_id": "msg20",
142   "user_id": "user1"
143 },
144 {
145   "amount": 1,
146   "message": "msg21",
147   "timestamp": 1774546788.488648,
148   "transaction_id": "msg21",
149   "user_id": "user1"
150 },
151 {
152   "amount": 1,
153   "message": "msg22",
154   "timestamp": 1774546794.6055605,
155   "transaction_id": "msg22",
156   "user_id": "user1"
157 }
158 ]
159 }
```

По постману бачимо що вони виконалися, але перевіримо ще логи logging service:


```

akravchuk@macA apz-kravchuk % docker logs apz-kravchuk-logging-service-1-1

[logging-service:1] stored tx=msg10 user=user1 amount=1 msg=msg10
172.18.0.10 - - [26/Mar/2026:17:07:18 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
172.18.0.10 - - [26/Mar/2026:17:09:04 +0000] "GET /transactions/user/user1 HTTP/1.1" 200 1049 "-" "python-requests/2.32.3"
[logging-service:1] stored tx=msg12 user=user1 amount=1 msg=msg12
172.18.0.10 - - [26/Mar/2026:17:28:33 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
172.18.0.10 - - [26/Mar/2026:17:30:10 +0000] "GET /transactions/user/user1 HTTP/1.1" 200 1675 "-" "python-requests/2.32.3"
[logging-service:1] stored tx=msg17 user=user1 amount=1 msg=msg17
172.18.0.10 - - [26/Mar/2026:17:39:27 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[logging-service:1] stored tx=msg18 user=user1 amount=1 msg=msg18
172.18.0.10 - - [26/Mar/2026:17:39:32 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[logging-service:1] stored tx=msg22 user=user1 amount=1 msg=msg22
172.18.0.10 - - [26/Mar/2026:17:39:54 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
akravchuk@macA apz-kravchuk % docker logs apz-kravchuk-logging-service-2-1

[2026-03-26 17:05:19 +0000] [1] [INFO] Starting gunicorn 22.0.0
[2026-03-26 17:05:19 +0000] [1] [INFO] Listening at: http://0.0.0.0:8001 (1)
[2026-03-26 17:05:19 +0000] [1] [INFO] Using worker: sync
[2026-03-26 17:05:19 +0000] [7] [INFO] Booting worker with pid: 7
[logging-service:2] stored tx=msg1 user=user1 amount=1 msg=msg1
172.18.0.10 - - [26/Mar/2026:17:05:42 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[logging-service:2] stored tx=msg3 user=user1 amount=1 msg=msg3
172.18.0.10 - - [26/Mar/2026:17:06:25 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[logging-service:2] stored tx=msg4 user=user1 amount=1 msg=msg4
172.18.0.10 - - [26/Mar/2026:17:06:32 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[logging-service:2] stored tx=msg6 user=user1 amount=1 msg=msg6
172.18.0.10 - - [26/Mar/2026:17:06:50 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[logging-service:2] stored tx=msg7 user=user1 amount=1 msg=msg7
172.18.0.10 - - [26/Mar/2026:17:06:57 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[logging-service:2] stored tx=msg8 user=user1 amount=1 msg=msg8
172.18.0.10 - - [26/Mar/2026:17:07:04 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[2026-03-26 17:26:55 +0000] [1] [INFO] Handling signal: term
[2026-03-26 17:26:55 +0000] [7] [INFO] Worker exiting (pid: 7)
[2026-03-26 17:26:55 +0000] [1] [INFO] Shutting down: Master
[2026-03-26 17:36:17 +0000] [1] [INFO] Starting gunicorn 22.0.0
[2026-03-26 17:36:17 +0000] [1] [INFO] Listening at: http://0.0.0.0:8001 (1)
[2026-03-26 17:36:17 +0000] [1] [INFO] Using worker: sync
[2026-03-26 17:36:17 +0000] [7] [INFO] Booting worker with pid: 7
[logging-service:2] stored tx=msg20 user=user1 amount=1 msg=msg20
172.18.0.10 - - [26/Mar/2026:17:39:43 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[logging-service:2] stored tx=msg21 user=user1 amount=1 msg=msg21
172.18.0.10 - - [26/Mar/2026:17:39:48 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
akravchuk@macA apz-kravchuk % docker logs apz-kravchuk-logging-service-3-1

[2026-03-26 17:05:19 +0000] [1] [INFO] Starting gunicorn 22.0.0
[2026-03-26 17:05:19 +0000] [1] [INFO] Listening at: http://0.0.0.0:8001 (1)
[2026-03-26 17:05:19 +0000] [1] [INFO] Using worker: sync
[2026-03-26 17:05:19 +0000] [7] [INFO] Booting worker with pid: 7
[logging-service:3] stored tx=msg11 user=user1 amount=1 msg=msg11
172.18.0.10 - - [26/Mar/2026:17:28:27 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[logging-service:3] stored tx=msg13 user=user1 amount=1 msg=msg13
172.18.0.10 - - [26/Mar/2026:17:28:37 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[logging-service:3] stored tx=msg14 user=user1 amount=1 msg=msg14
172.18.0.10 - - [26/Mar/2026:17:28:43 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[logging-service:3] stored tx=msg15 user=user1 amount=1 msg=msg15
172.18.0.10 - - [26/Mar/2026:17:28:48 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[logging-service:3] stored tx=msg16 user=user1 amount=1 msg=msg16
172.18.0.10 - - [26/Mar/2026:17:28:56 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
[logging-service:3] stored tx=msg19 user=user1 amount=1 msg=msg19
172.18.0.10 - - [26/Mar/2026:17:39:36 +0000] "POST /transactions HTTP/1.1" 200 12 "-" "python-requests/2.32.3"
akravchuk@macA apz-kravchuk %

```

Бачимо що сервіс 1 отримав транзакції 17, 18 і 22, сервіс 2 отримав транзакції 20 і 21, а сервіс 3 отримав транзакцію 19. Все чудово працює навіть без однієї ноди.

Тепер можемо провести тести як в першому завданні:

```
akravchuk@macA apz-kravchuk % python3 tests.py

=== Scenario 1 (10 clients -> 10 accounts) ===
Total requests: 100000
Total time (sec): 367.890
Requests/sec: 271.8
facade.metrics.logging_calls=24981 total_logging_time_sec=229.885 avg_ms=9.20 share_of_wall=62.5%
facade.metrics.counter_calls=24981 total_counter_time_sec=188.949 avg_ms=7.56 share_of_wall=51.4%
Scenario 1 correctness OK: 10 users have balance=10000

=== Scenario 2 (10 clients -> 1 shared account) ===
Total requests: 100000
Total time (sec): 369.005
Requests/sec: 271.0
facade.metrics.logging_calls=24955 total_logging_time_sec=233.519 avg_ms=9.36 share_of_wall=63.3%
facade.metrics.counter_calls=24955 total_counter_time_sec=194.716 avg_ms=7.80 share_of_wall=52.8%
Scenario 2 correctness OK: shared-1774724740 has balance=100000
```

У першому сценарії, де 10 користувачів виконували по 10 000 транзакцій кожен на власний рахунок, система обробила 100 000 запитів за 367.89 секунд, що становить приблизно 272 запити на секунду. Середній час виклику logging-service склав близько 9.20 мс, а counter-service — близько 7.56 мс. У порівнянні з першим завданням продуктивність знизилась, що пояснюється додатковими накладними витратами на мережеву взаємодію та використання розподіленого сховища Hazelcast. Проте система залишається стабільною та коректно обробляє паралельні запити.

У другому сценарії всі 100 000 транзакцій виконувалися на один спільний рахунок. Загальний час виконання становив 369.01 секунд, а продуктивність склала приблизно 271 запит на секунду. Незначне зниження швидкості у порівнянні з першим сценарієм пов'язане з конкуренцією за доступ до одного ресурсу в counter-service. При цьому середній час викликів залишився майже незмінним, що підтверджує коректну синхронізацію та відсутність race condition.

Загалом, у порівнянні з першим завданням спостерігається зниження продуктивності через використання Hazelcast та розподіленої архітектури, однак система стала більш масштабованою та відмовостійкою, зберігаючи стабільність роботи під навантаженням.